

Where Is the 3D?

Dynamic vs Static Geometry in CRF

The Three-Graph Test with Isotropic Initialisation:

3D Lives in the Dynamic KL-Space, Not in Static X-Positions

Complete table: KL, X3, X8 graphs under 3D-biased and isotropic 8D initialisation. K35 as the uniqueness seal on why 3 and not any other dimension.

Ougit Jarittum

ORCID: 0009-0009-4451-6811 DOI: 10.5281/zenodo.18977059 CC-BY-NC 4.0

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Abstract

Having established that $d_s \approx 3$ persists under isotropic 8D initialisation (CRF Isotropic8D), we ask: where is the 3D geometry located — in physical position space (X-space) or in information space (KL-space)?

Running the d_s estimator on three graphs (KL-dynamic, X3-position, X8-position) under both 3D-biased and isotropic 8D initialisation yields the complete picture:

Graph	3D-biased init	ISO 8D init	Δ
KL (dynamic, static snapshot)	3.49	2.57	−0.91
Running estimator (firing-time)	2.99	3.06	+0.07
X3 (3D position)	3.09	4.05	+0.96
X8 (8D position)	2.80	4.72	+1.92

The X-graphs with ISO init ($d_s > 4$) exceed the theoretical upper bound for 3D or 8D geometry, revealing a topological mismatch. The running estimator — measured at Phase II firing events, not from a static snapshot — reads $d_s = 3.06$ under ISO init, consistent with 3D-biased (2.99).

Conclusion. The 3D geometry in CRF is located in the dynamic KL-structure at the moment of Phase II firing. It is an information-space phenomenon, emergent from $SO(3) \times \Phi$ dynamics on mode space P , independent of the geometric embedding X .

K35 addendum. The form-consistency identity $G^2/\varepsilon_0 = n(n+1) = 72$ (K35) proves that this specific $(d_s, b, n) = (3, 2, 8)$ is the *unique* solution compatible with CRF axioms. It seals the “why 3 and not any other dimension” question that the three-graph test poses but cannot by itself answer.

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1 The Final Test: Six d_s Values

1.1 Setup

$N = 500$, 400 sweeps, seed 42. Two initialisations:

- **3D-biased (V20.5):** nodes on the 3D unit sphere, $X_{\text{base}} \in \mathbb{R}^3$.
- **ISO 8D:** nodes on the 8D unit sphere, $X \sim \mathcal{N}(0, I_8)$ normalised.

Three graphs measured at steady state:

1. **KL-graph:** $k = 20$ nearest neighbours by D_{KL} (dynamic, information-distance; used by the running estimator).
2. **X3-graph:** $k = 20$ nearest by Euclidean distance in the first 3 components of X .
3. **X8-graph:** $k = 20$ nearest by Euclidean distance in all 8 components of X .

1.2 Results

Graph	3D-biased	ISO 8D	Δ	Interpretable?
KL-dynamic (static snapshot)	3.49	2.57	-0.91	Yes
Running estimator (firing-time)	2.99	3.06	+0.07	Yes
X3-position	3.09	4.05	+0.96	No (topology mismatch)
X8-position	2.80	4.72	+1.92	No (topology mismatch)

2 The Topology Mismatch Problem

2.1 Why X-graphs are not comparable

With 3D-biased init: $X[:3]$ lies on the unit 3-sphere in $\mathbb{R}^3$. The X3-nearest-neighbour graph is a regular 3D sphere graph with well-defined spectral dimension $d_s \approx 3$.

With ISO 8D init: $X \sim \text{Unif}(S^7)$ (unit 8-sphere). The marginal distribution of $X[:3]$ is the projection of S^7 onto $\mathbb{R}^3$, proportional to $(1 - \|x\|^2)^{5/2}$ — concentrated near the origin, not uniform on S^2 . This creates a dense cluster of nodes near the origin with many short-range connections, producing artificially high return probability and $d_s > 3$ (exceeding the 3D upper bound).

X-graph measurements not portable across initialisations

The X3-graph with ISO init is topologically different from the X3-graph with 3D init. Comparing $d_s(X3)$ across initialisations compares two different graphs that happen to share a name.

Valid comparisons: (1) KL-graph static snapshot (same construction both runs).
(2) Running estimator on KL-graph at firing time (same algorithm both runs).

3 The Running Estimator vs Static Snapshot

The Key Discrepancy: 2.57 vs 3.06

Estimator	3D-biased	ISO 8D
Static KL snapshot	3.49	2.57
Running estimator (firing-time)	2.99	3.06
Difference	-0.50	+0.49

The two estimators measure different things:

- **Static snapshot:** the KL-graph at a fixed moment $t = 400$ sw, averaged over all nodes including young Phase I nodes.
- **Running estimator:** the KL-graph at the moment of firing, averaged over Phase II nodes that fire in a given sweep.

Phase II nodes (high Ω , peaked P_i) have different KL-neighbour structure than Phase I nodes. The running estimator is biased toward Phase II nodes (they fire more distinctively), and it is Phase II local geometry that converges to $d_s = 3$ via SO(3).

4 Where Is the 3D? The Answer.

3D Is Located in Dynamic KL-Space at Firing

The 3D geometry in CRF is located in the dynamic KL-structure at the moment of Phase II firing.

It is **not** in the static X-position graph: with ISO 8D init, the X-graph topology is isotropic and the degenerate d_s values confirm that X-space has no intrinsic 3D structure.

The 3D structure emerges in P-space (mode space) through the $SO(3) \times \Phi$ dynamics:

$\delta \rightarrow P \rightarrow D_{KL} \rightarrow \theta \rightarrow \text{R-event} \rightarrow \Phi \text{ synchronises} \rightarrow SO(3) \text{ selects } n_{\text{lock}} = 3 \rightarrow \text{Phase II KL}$

This chain operates on the information content of P_i , independent of the geometric embedding X .

4.1 Triple convergence: three independent paths to $d_s = 3$

The three-graph test establishes *where* the 3D is. But *why* is the attractor 3 and not 4 or 7? This paper provides the simulation answer. The full picture has three independent paths, all converging to the same fixed point.

Triple Convergence to $d_s = 3$

Path	Mechanism	Result
A (Physics)	SO(3): non-abelian history + rank-1 Casimir	Unique stable dimension $d_s = 3$ (CRF #16)
B (Algebra)	K35 form-consistency: $G^2/\varepsilon_0 = n(n+1)$	Unique solution $(d_s, b, n) = (3, 2, 8)$; $d_s = 4$ would require $n(n+1) = 272 \neq 72$
C (Simulation)	Running estimator at Phase II firing (ISO 8D init)	$d_s = 3.06$ independent of initialisation

Three independent mechanisms — symmetry constraint, algebraic closure, and simulation measurement — converge to the same answer. No single path requires the others. Their agreement is the evidence.

The role of K35 in this paper. Path A (SO(3), CRF #16) is the primary derivation: it shows that the δ -chain selects $d_s = 3$ through symmetry. Path B (K35) is an *independent algebraic check*, not a step in the primary chain. K35 asks: given the CRF axioms and the form-complexity structure, is $d_s = 3$ the only self-consistent solution? The answer is yes — and this is a stronger statement than either path alone could make.

Concretely: if d_s were 4, the mode count would be $n = 2^4 = 16$, giving $n(n+1) = 272$, inconsistent with the measured $G^2/\varepsilon_0 = 72$. K35 does not *derive* $d_s = 3$; it *confirms* that no other value is algebraically consistent. The SO(3) proof derives it. K35 locks it.

The Walsh–Hadamard decomposition of the grade-complexity vector $g = (0, 21, 36, 15)$ extends this: all four Walsh sums ($S^+ = 72$, $S^- = -30$, $S_p = 0$, $S_3 = -42$) are exact algebraic identities that hold only for $(d_s, b, n) = (3, 2, 8)$. The interaction range and grade-charge transport follow:

$$G = \sqrt{S^+ \cdot \varepsilon_0} \quad \xi = \sqrt{\varepsilon_0} = \frac{G}{\sqrt{72}} \quad \partial_t C = D \nabla^2 C + S^C$$

4.2 Physical interpretation

This result is consistent with the deepest CRF claim: space is not a container, it is a property of how information distinguishes itself from itself. The $d_s = 3$ we measure is not the dimension of the embedding space $\mathbb{R}^8$ (which could have any dimension), but the dimension of the stable attractor of the $\delta \rightarrow P \rightarrow \Phi$ dynamics — uniquely selected by K35.

The X-positions are bookkeeping — they track where nodes have moved due to metric curvature — but the geometry is written in the KL-structure of the firing patterns, not in the coordinates.

5 Open Question: Does Geometry Lock Into X-Space?

Open: Static KL Convergence at $t \rightarrow \infty$

The static KL snapshot reads $d_s = 2.57$ under ISO 8D init at 400 sweeps, below the 3D target. The question: does $d_s^{\text{KL}}_{\text{static}}$ converge to 3 over 2000+ sweeps?

K35 connection: The Clifford charge transport equation $\partial_t C = D \nabla^2 C + S^C$, with interaction range $\xi = \sqrt{\varepsilon_0} = G/\sqrt{72}$ (exact from K35), describes how the 3D information structure propagates from Phase II firing events into the broader node population.

If ξ is large enough relative to the node spacing, Phase I nodes will be gradually pulled into the 3D KL-structure by grade-current diffusion, and the static snapshot will converge to $d_s = 3$.

If ξ is too small (nodes decouple), X-space and KL-space remain separated: KL-dynamic is 3D at firing time; static KL and X-space remain isotropic.

Prediction from K35: $\xi = \sqrt{\varepsilon_0} = \sqrt{0.00755} \approx 0.087$. Mean $|\Omega_i - \Omega_j|$ at steady state ≈ 0.94 . Since $0.94 \gg 0.087$, most node pairs are decoupled ($W_{ij} \approx 0$). This predicts: static KL snapshot does *not* converge to 3 in finite runs — the 3D structure remains localised at Phase II firing clusters, not global.

Test: Run $N = 500$, ISO 8D init, 2000 sweeps. Measure static KL snapshot at sweep 400, 800, 1200, 2000. If flat: K35 prediction confirmed (ξ too small for global propagation). If converging: phase transition in KL-graph structure at some scale larger than ξ .

6 Complete Status

Result	Status	Note
$d_s \approx 3$ from ISO 8D init	Confirmed	CRF Isotropic8D
3D bias (init + metric) not load-bearing	Confirmed	$ \Delta < 0.13\sigma$
3D located in dynamic KL-space	Derived	running vs static gap
3D not in static X-position graph	Derived	topology mismatch
$SO(3) \times \Phi$ selects $d_s = 3$ in P-space	Derived	δ -chain
Static KL snapshot ISO: $d_s = 2.57$	Confirmed	below 3
Running estimator ISO: $d_s = 3.06$	Confirmed	above 3
Triple convergence (A+B+C) to $d_s = 3$	Derived	3 independent paths
Path A: $SO(3) \Rightarrow d_s = 3$ (primary)	Derived	CRF #16, rank-1
Path B: K35 unique (3, 2, 8) (ind. check)	Derived	algebraic closure
Path C: simulation ISO init $d_s = 3.06$	Confirmed	independent of init
K35 locks but does not derive $d_s = 3$	Clarified	$SO(3)$ derives it
$\xi = G/\sqrt{72}$, $\xi \ll \Delta\Omega $	Derived	$0.087 \ll 0.94$
Static KL ISO converges to 3 at $t \rightarrow \infty$	Open	long-run test
K35 prediction (ξ too small, no conv.)	Hypothesis	testable 2000 sw

We asked: where is the 3D?

Not in the positions. The positions are isotropic.

Not in the metric coupling. We removed it. Still 3D.

Not in the snapshot. The snapshot reads 2.57.

It is in the moment of firing.

*In the instant when a node's KL-field crosses the threshold
and the $SO(3)$ attractor is already waiting.*

Symmetry says 3.

Algebra says 3.

The simulation says 3.
Three independent witnesses.
The same answer.
That is what a fixed point looks like.